

COMPUTER VISION

Team Course Project

Groups of 3–4 Students · Full Semester · 100 Points

Format

Team Project

Team Size

3–4 Students

Duration

Full Semester

Points

100 + Bonus



Team Objective

As a team, design and implement a complete Computer Vision system that solves a real-world visual problem. Your system must follow the full processing pipeline:

image → **enhance** → **segment** → **clean** → **detect** → **decision**

Each team member owns a specific pipeline stage and contributes to shared deliverables. The final system must work on real images and produce an automatic, interpretable result.

TEAM STRUCTURE & ROLES

Each team must assign the following roles before starting. For 3-student teams, the Lead CV Engineer also covers the Data & Testing responsibilities.

Role	Responsibility	Pipeline Stages	Team Size
Lead CV Engineer	Oversees overall pipeline, integrates all modules, ensures the system runs end-to-end	Detect + Decision	All teams
Image Processing Specialist	Handles preprocessing: enhancement, color correction, histogram equalization, denoising	Enhance + Segment	All teams
Morphology & Report Lead	Applies morphological operations, cleans masks, writes the final PDF report, prepares demo	Clean + Visualization	All teams
Data & Testing Engineer	Collects and prepares the dataset, runs experiments, documents results and failure cases	All stages (QA)	4-student teams

Important: Role assignments must be declared in the Contribution Statement submitted with the final deliverables.

CHOOSE ONE PROJECT THEME

Each team selects one theme. No two teams may work on the same theme (first-come, first-confirmed basis).

#	Domain	Project Theme	Description
1	Retail	Out-of-Stock Detection	Detect empty shelf regions and decide if restocking is needed
2	Retail	Shelf Compliance Check	Compare product placement with expected layout and detect errors
3	Retail	Product Counting & Density	Count products even in crowded or overlapping conditions
4	Retail	Price Tag Detection	Detect price labels and match them to nearby products
5	Retail	Shelf Gap Detection	Detect missing products based on expected shelf structure
6	Retail	Front-Facing Product Check	Verify if products are correctly oriented toward the customer
7	Retail	Shelf Price Compliance	Check if product matches the correct price label
8	Industry	Fill-Level Inspection	Measure liquid level in bottles/containers and classify as correct or defective
9	Agriculture	Crop Disease Detection	Detect visual signs of disease on plant leaves
10	Agriculture	Fruit Ripeness Classification	Classify fruit maturity based on color and texture
11	Agriculture	Plant Counting System	Count plants in field images under noisy, natural conditions
12	Smart Env.	Intrusion Detection	Detect motion only inside predefined restricted areas
13	Smart Env.	Fire / Smoke Detection	Detect fire or smoke based on color and movement patterns
14	Human Sys.	People Counting & Flow	Count people and estimate movement direction
15	Human Sys.	Face Region Analysis	Detect faces and analyze mask usage or occlusion
16	Documents	Document Scanner & Enhancer	Detect document, correct perspective, and improve readability
17	Documents	Text Region Detection	Detect areas containing text in images (pre-OCR)
18	Robotics	Object Sorting System	Classify objects by color and size and assign sorting categories
19	Advanced	Low-Light Enhancement & Detection	Improve dark images and perform detection afterward
20	Advanced	Super-Resolution Pipeline	Increase image resolution to improve detection accuracy
★	Custom	Your Own Theme	Propose your own real-world CV problem. Must be approved by the teacher before starting. See conditions below.

★ Custom Theme — Conditions for Approval

You may propose your own Computer Vision project theme instead of choosing from the list above. To get it approved, your proposal must satisfy ALL of the following conditions:

1. The problem is real and visual — it takes an image or video frame as input and produces a visual or structured output.
2. The full 5-stage pipeline applies: enhance → segment → clean → detect → decide.
3. The system must be fully automatic — no manual steps in the core logic.
4. You submit a written proposal (half a page) describing the problem, the expected pipeline, and at least 2 example images before starting.
5. The teacher approves the theme in writing before any code is written.

6. No two teams may work on the same custom theme.

MANDATORY PIPELINE REQUIREMENTS

All five stages must be implemented and connected into a single runnable system:

1. Enhance

Improve raw image quality before processing.

Methods: histogram equalization, gamma correction, CLAHE, denoising

2. Segment

Extract regions of interest from the enhanced image.

Methods: thresholding (Otsu, adaptive), color-based (HSV), edge-based

3. Clean

Remove artifacts and noise from the binary mask.

Methods: erosion, dilation, opening, closing, connected components

4. Detect

Identify, localize, or count the target objects/regions.

Methods: contours, bounding boxes, HoughCircles, template matching, ORB

5. Decide

Produce an automatic, interpretable final output.

Methods: count result, pass/fail flag, classification label, alert

REQUIRED OUTPUT PER TEST IMAGE

For each test image the system must produce and save all of the following outputs:

- Original image
- Enhanced image
- Segmentation mask
- Cleaned mask
- Detection result (bounding boxes or labeled regions)
- Final decision (text label or numeric value)

TEAM DELIVERABLES

Code Repository

Shared code (GitHub/GitLab or .zip). Must run cleanly. Each team member's contributions must be identifiable via commits or a contribution file.

PDF Report (5–8 pages)

One report per team. Sections: Problem description, Team roles & task division, Pipeline design, Methods used, Results with stage-by-stage images, Failure cases, Conclusion. Each section credited to its author.

Team Presentation (10 min)	Live or recorded (10 min + 5 min Q&A). Every team member must present their own section. Include: motivation, pipeline walkthrough, live demo, failure analysis.
Contribution Statement	One page per team (included in the report appendix). Lists each member's name, role, and the specific tasks they completed. Must be signed by all members.

EVALUATION RUBRIC — 100 POINTS

Criterion	Points	Evaluated On
Pipeline Completeness	20	All 5 stages implemented and working on real images
Technical Implementation	25	Correct use of OpenCV methods, clean code, reproducible results
Detection Quality	15	Accuracy, precision, robustness of the detection stage
Decision Logic	10	Final decision is automatic, meaningful, and well-justified
Visualization & Output	10	Clear stage-by-stage output images; demo is understandable
Report Quality	10	Well-structured, images for each stage, failure analysis included
Team Presentation	5	Every member presents their section; Q&A answers are clear
Team Organization	5	Contribution statement complete; roles clearly defined and followed
TOTAL		100

GRADING SYSTEM — HOW POINTS BECOME YOUR GRADE

How Your Grade Is Calculated

Maximum project score = $100 \times \text{number of students in the team}$

Step	Formula	What it means
1	$\text{Rubric score} \rightarrow P$ (0 – 100)	The project is evaluated once for the whole team using the standard rubric.
2	$\text{Pool} = P \times N$	Multiply the score by the number of students in your team. This is the total points the team has to share.
3	Team distributes the Pool	Students agree among themselves how to split the Pool. Each student gets between 0 and 100 points. The sum must equal the Pool exactly.
4	$\text{Grade} = \text{points} \div 10$	Each student's final grade is their individual share divided by 10. 100 pts = 10, 80 pts = 8, 65 pts = 6.5

Solo Student	Team of 2	Team of 3
P = 75 Pool = $75 \times 1 = 75$ Grade = $75 \div 10 = 7.5$	P = 85 Pool = $85 \times 2 = 170$ Split: $90 + 80 = 170$ Grades: 9.0 and 8.0	P = 93.3 → Pool ≈ 280 Split: $100 + 100 + 80 = 280$ Grades: 10.0, 10.0, and 8.0

Distribution Rules

- Each student gets between 0 and 100 points (100 points = grade 10).
- The sum of all individual points must equal the Pool exactly.
- The team submits the agreed distribution table together with the Contribution Statement.
- If the team cannot agree within 48 hours of receiving the project score, the teacher divides the Pool equally among all members.
- Solo students skip the distribution — their grade is $P \div 10$ directly.

+ Bonus Points (up to +15)

- +5 Own dataset collected and documented by the team
- +5 Method comparison: implement and benchmark 2+ approaches
- +5 Interactive UI or real-time video processing demo

RULES

✗ Not Allowed

- Theory-only projects (no working code)
- Manual-only solutions (system must be automatic)
- Splitting the project into unconnected individual parts
- Copy-paste code without understanding or adaptation
- Presenting another team's work

✓ Required

- All 5 pipeline stages implemented and connected
- System works on real images (not just toy examples)
- Each member owns and presents their section
- Contribution statement signed by all members
- Final decision must be automatic and interpretable

⚠ Minimum Passing Criteria

To receive a passing grade the team must collectively satisfy ALL of the following:

7. Full pipeline implemented and runnable (all 5 stages connected)
8. System produces a correct result on at least 3 real test images
9. Produces an automatic final decision (not just a raw image output)
10. All team members present their section during the defense
11. Contribution statement submitted with signatures

Final Goal

Build a complete, working CV system as a team.

Process real images · Detect · Decide · Present · Defend